

- Q.26 Write short note on slip casting.
 Q.27 Define water of plasticity.
 Q.28 Draw alumina-Silica phase diagram and explain it.
 Q.29 Explain the terms hardness and brittleness.
 Q.30 Draw the diagram of water system.
 Q.31 Explain Gibbs phase rule.
 Q.32 Explain magnetizing force.
 Q.33 Differentiate between creep and fatigue failure.
 Q.34 Discuss thermal properties of materials.
 Q.35 Differentiate between hard and soft magnets.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
 Q.36 Explain the theory of valance with arrangement of electrons of **s, p, d** and **f** orbits.
 Q.37 Explain the point, fine and surface defects.
 Q.38 Explain Electrical properties of ceramic product.

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3rd Sem / Ceramic Sub : Ceramic Science

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 _____ bond is formed by sharing of electrons between two atoms.
 a) Ionic Bonding b) Hydrogen Bonding
 c) Covalent Bonding d) None of the above
- Q.2 In _____ crystal system atoms occupies corner.
 a) Simple Cubic b) Body Centered
 c) Hexagonal d) Face Centered
- Q.3 The following is a type of crystal defect
 a) Interstitiality b) Vacancy
 c) Screw dislocation d) All of the above
- Q.4 Chemical formula of Kaolin is represented by _____.
 a) $\text{Al}_2\text{O}_3\text{SiO}_2$ b) $\text{Al}_2\text{O}_3\cdot 2\text{SiO}_2\cdot 2\text{H}_2\text{O}$
 c) $\text{Al}_2\text{O}_3\cdot 2\text{SiO}_2$ d) $\text{Al}_2\text{O}_3\cdot 2\text{SiO}_2\cdot \text{H}_2\text{O}$
- Q.5 The montmorillonite clay mineral is highly _____.
 a) Elastic b) Plastic
 c) Hard d) Tough

- Q.6 The failure of a material due to cyclic or reversing loads is called _____.
 a) Creep b) Fatigue
 c) Brittle failure d) Ductile failure
- Q.7 Which material is more ductile?
 a) Steel b) Rubber
 c) Plastic d) Ceramic
- Q.8 In phase diagrams, the material _____ solidus line is solid.
 a) Above b) Below
 c) On the line d) None of the above
- Q.9 In phase diagrams we can study _____.
 a) Mixing behavior b) Firing behavior
 c) Drying behavior d) None of the above
- Q.10 The number of magnetic lines of force set up in a magnetic circuit is called _____.
 a) Magnetic flux b) Flux density
 c) Tesla d) All of the above

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Sharing of electrons between two atoms creates a _____ bond. (Ionic/Covalent)

- Q.12 The chemical formula for alumina is _____.
 Q.13 Normally the colour of Kaolinite is _____. (White/Yellow)
 Q.14 The example of point defect is cracking. (True / False)
 Q.15 The clay shows plasticity only when _____ is added. (Water / Grease)
 Q.16 The material which regains its shape after removal of load is called plastic. (T/F)
 Q.17 Example of 2 component phase diagram is water system. (True/False)
 Q.18 Which magnets have higher coercive force value? (Soft/hard)
 Q.19 Soft magnetic materials can be easily demagnetized. (True/False)
 Q.20 Alumina-Silica form binary phase diagrams? (True/False)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Write short note on ionic bond with example.
 Q.22 Explain space lattice and unit cell.
 Q.23 Write short note on Electronic configuration of atoms.
 Q.24 Explain any two physical characteristics of clay.
 Q.25 Write short note on clay chemistry.